

RE016 – AI Service Selection and Cost Sustainability

Review

Summary

Food Remedy should not make AI responsible for core food-safety or recommendation decisions. Existing Food Remedy research recommends a deterministic, safety-first recommendation pipeline in which allergies, dietary restrictions, medical restrictions, nutritional calculations, scoring and ranking are handled by rules rather than an AI model. Structured/template explanations can also operate without AI

The most appropriate use of AI is therefore as an optional explanation and personalization layer after a safe recommendation has already been produced. This allows Food Remedy to gain the benefits of natural-language explanations without allowing an LLM to override safety decisions.

Three AI deployment approaches were reviewed:

- **Hosted API:** the most practical option for Food Remedy because it requires little infrastructure and can scale with usage.
- **Self-hosted/open-weight model:** possible, but requires hosting, maintenance and computing resources.
- **On-device AI:** can remove API costs and improve privacy but introduces device compatibility and cross-platform development requirements.

For the current Food Remedy requirements, **Google Gemini 2.5 Flash-Lite is recommended as the preferred paid hosted model** because of its low token cost and available free tier. Its current standard price is US\$0.10 per million input tokens and US\$0.40 per million output tokens.

However, because there will be no ongoing funding after release, the application must remain fully functional without an AI service. The recommended no-cost solution is the existing deterministic recommendation engine combined with structured explanation templates.

Background and Food Remedy Requirements

Food Remedy's recommendation research identifies a safety-first recommendation architecture. The proposed pipeline collects user, food and contextual information, applies strict safety rules, calculates nutritional progress, scores eligible foods and ranks them before presenting recommendations.

The recommendation of architecture specifically identifies the following information:

- user allergies, diets, medical restrictions and preferences
- food ingredients, allergens, dietary tags, nutrients, price and preparation time
- health constraints and validated rules
- nutritional goals
- contextual information
- rule results
- candidate scores
- recommendation results

This information is sufficient to generate a recommendation without an LLM.

AI requirement

The AI service should therefore be treated as an enhancement rather than a core safety component.

Potential AI functions include:

1. explaining why an already-validated product was recommended
2. presenting recommendation information in natural language
3. personalizing the explanation based on the user's preferences
4. comparing already-approved products in a user-friendly way

The AI must **not**:

- determine whether a food is safe for an allergy
- override dietary restrictions
- Making Medical Decisions
- independently select unsafe products
- replace deterministic recommendation rules
- become necessary for the basic operation of the application

RE008 explicitly states that machine learning should only operate within the soft ranking/personalization layer and must not override the deterministic safety rules.

2. Suitable AI Deployment Options

2.1 Hosted AI API

A hosted API means Food Remedy sends a controlled request to an external AI provider and receives the generated explanation.

Advantages

- No model training is required.
- No GPU infrastructure required.
- Easy to scale.
- Suitable for short explanation-generation requests.
- Pricing is based on actual usage.
- Model can be replaced later without redesigning the whole recommendation engine.

Disadvantages

- Requires internet connectivity.
- Creates an ongoing usage cost if the paid service is used.
- User information sent to the provider requires privacy controls.
- The feature can become unavailable if the API account has no remaining credits.

For Food Remedy, the hosted API approach is the most practical AI deployment option because the intended AI workload is small and non-authoritative.

2.2 Self-hosted/open-weight model

An open-weight model can be deployed on infrastructure controlled by Food Remedy.

This can reduce dependence on a commercial API and provide greater control over data. However, "free model" does not mean "zero cost". The project would still need computing resources, storage, deployment, monitoring, maintenance and potentially GPU resources.

Open-weight models are therefore more attractive if Food Remedy eventually reaches sufficient scale to justify its own infrastructure.

For the current project, self-hosting would introduce unnecessary operational complexity.

2.3 On-device AI

On-device AI runs the model directly on the user's device.

Modern mobile platforms support on-device language models. For example, Apple's Foundation Models framework provides on-device language-model functionality and supports text generation and structured tasks, but availability depends on compatible devices and Apple Intelligence support. Apple also documents Core AI as providing on-device model execution without server dependencies or token costs.

However, Food Remedy uses Expo/React Native, so building a cross-platform on-device AI implementation would require additional platform-specific development. It would also need to handle devices that do not support the required model. Therefore, on-device AI is technically interesting but not the preferred approach for the current release.

3. AI Provider Comparison

The following providers were compared using their current official API pricing.

Provider/model	Input / 1M tokens	Output / 1M tokens	Free option	Suitability
Google Gemini 2.5 Flash-Lite	US\$0.10	US\$0.40	Yes, subject to limits	High
OpenAI GPT-5 mini	US\$0.25	US\$2.00	Account-dependent credits	High
Claude Haiku 4.5	US\$1.00	US\$5.00	No comparable permanent API free tier identified	Medium

Google describes Gemini 2.5 Flash-Lite as a small, cost-effective model designed for at-scale usage. Its standard paid pricing is US\$0.10 per million input tokens and US\$0.40 per million output tokens, while its free tier is subject to usage limits.

OpenAI's GPT-5 mini is priced at US\$0.25 per million input tokens and US\$2.00 per million output tokens. OpenAI's API also uses prepaid billing for new accounts; if an account has free credits, those credits are used before purchased credits. When the available balance is exhausted, API requests can return a billing error.

Anthropic's Claude Haiku 4.5 is priced at US\$1 per million input tokens and US\$5 per million output tokens.

4. Tokenization and Cost Model

AI API pricing is generally based on tokens. A token is a unit of text processed by the model. API costs are calculated separately for input tokens and output tokens.

For planning purposes, the following Food Remedy request is assumed:

- 800 input tokens per request
- 300 output tokens per request
- 1,100 tokens total per request

This is a planning estimate. Actual Food Remedy token usage should be measured during prototype testing.

The monthly cost can be estimated using: **Monthly cost = (Input tokens ÷ 1,000,000 × input price) + (Output tokens ÷ 1,000,000 × output price)**

5. Estimated Food Remedy Monthly Cost

Three usage scenarios were modelled.

Scenario	AI requests/month	Input tokens	Output tokens
Low	5,000	4 million	1.5 million
Expected	20,000	16 million	6 million
High	100,000	80 million	30 million

Estimated API cost

Provider	Low	Expected	High
Gemini 2.5 Flash-Lite	US\$1.00	US\$4.00	US\$20.00
GPT-5 mini	US\$4.00	US\$16.00	US\$80.00
Claude Haiku 4.5	US\$11.50	US\$46.00	US\$230.00

These estimates represent AI token charges only. They do not include Firebase, backend hosting, database, networking or other application costs. The calculations demonstrate that the limited explanation/personalization workload is relatively inexpensive when a low-cost model is used.

6. Free and Cost-Efficient Alternatives

Gemini

Gemini API accounts currently begin on the Free Tier, which provides access to eligible models within their applicable free usage limits.

Gemini 2.5 Flash-Lite currently has free input and output pricing on the Free Tier.

This makes it suitable for:

- development;
- experimentation;
- prototype testing; and
- demonstrating the AI explanation concept.

However, Food Remedy should not design its production architecture around an assumed permanent free quota because free-tier availability and limits can change.

OpenAI credits

OpenAI's current prepaid billing documentation states that accounts may have free credits and that these are consumed before purchased credits. New accounts use prepaid billing, and purchased credits expire after one year.

Therefore, OpenAI credits can be useful for development and testing, but they should not be treated as a guaranteed permanent free production service.

Anthropic

Anthropic's API uses prepaid usage credits for most organizations. Once credits are exhausted, API calls cannot continue until additional credits are added. Anthropic also offers specific free-credit research programs, but these are targeted at particular research areas and therefore should not be treated as a general Food Remedy trial allowance.

8. Cost-Efficient and Non-AI Alternatives

Option 1 – Structured explanation templates

This is the strongest option for Food Remedy. Instead of generating an explanation with AI, the system can combine validated information into predefined sentences.

For example: "This product was recommended because it matches your dietary preferences and passed the applicable allergen and suitability checks."

This has:

- no AI cost;
- no external dependency;

- predictable output;
- simple testing;
- no token usage; and
- minimal additional privacy exposure.

RE008 already identifies structured/template explanations as an approach that can operate without ML.

Option 2 – Open-weight model

A small open-weight model can be run locally or on Food Remedy infrastructure. The model software may be available at no licensing cost, but computing infrastructure and maintenance still create costs.

Option 3 – On-device model

An appropriate on-device model could remove API charges completely, but device compatibility and cross-platform development make this more complex for the initial release.

9. Recommended Free Plan

Because no ongoing funding will be available after release, the **free/no-funding configuration should not depend on an external AI provider.**

Recommended free architecture

Food/product data

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Deterministic safety rules

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Nutrition calculations

↓

Scoring and ranking

↓

Validated recommendation

↓

Structured explanation template

↓

User

Cost

AI API cost: US\$0/month

The application can still provide its essential recommendation and explanation functionality without an AI subscription. A free AI tier such as Gemini can be used during research, development and testing where appropriate, but it should remain an optional component.

10. Recommended Paid Plan

The recommended paid provider is **Google Gemini 2.5 Flash-Lite**

It is the strongest fit because:

- the required AI task is relatively simple;
- the model is positioned for cost-efficient, high-scale use;
- the current input/output price is very low;
- a Free Tier is available;
- It avoids the infrastructure burden of self-hosting;
- It can be added as an optional backend service.

Currently paid pricing is US\$0.10 per million input tokens and US\$0.40 per million output tokens.

Using the expected scenario of 20,000 requests per month, the estimated model cost is approximately: **US\$4 per month**

This should be treated as an estimate rather than a guaranteed monthly bill because actual prompt size, output size and usage will vary.

11. What Happens if Funding Ends?

The system should use a graceful degradation strategy.

Normal funded operation

User → Food Remedy backend → deterministic safety/recommendation engine → Gemini → personalized explanation → user

If AI funding ends

User → Food Remedy backend → deterministic safety/recommendation engine → structured explanation → user

The AI layer is therefore optional.

The following functionality remains:

- allergen checking;
- dietary restriction enforcement;
- medical restriction rules;
- nutritional calculations;
- product scoring;
- recommendation ranking;
- safe product alternatives;
- structured explanations.

Only the AI-enhanced natural-language explanation/personalization would be lost.

This is consistent with RE008's recommendation that the deterministic safety gate remains mandatory and that AI/ML should not override safety decisions.

12. Recommended Integration

If Food Remedy implements the AI layer, the AI API should be accessed through the backend, rather than directly from the mobile application.

The proposed flow is:

React Native / Expo app

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Food Remedy backend

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Deterministic recommendation engine

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Validated Recommendation Result

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Optional AI explanation service

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Explanation returned to app

This keeps the AI provider's credentials away from the mobile application and allows the backend to control:

- which information is sent to the model;
- prompt structure;
- usage limits;
- error handling;
- fallback behavior; and
- switching between AI providers.

The existing recommendation architecture already proposes a backend service and structured recommendation result containing ranked foods and explanations.

12. Final Recommendation

The research indicates that Food Remedy should not make AI a requirement for its core recommendation or safety functionality. The recommended architecture is a hybrid-ready system:

Core system

Deterministic rules + transparent scoring + structured explanations

Optional AI system

Hosted LLM for natural-language explanation and personalization

Free/no-funding option

Deterministic recommendation engine + structured explanation templates

AI cost: US\$0/month

Paid option

Google Gemini 2.5 Flash-Lite

Current price:

- Input: US\$0.10 / 1M tokens
- Output: US\$0.40 / 1M tokens

Estimated expected usage: US\$4/month

Alternative

Open weight/self-hosted model can be considered later if usage becomes large enough to justify infrastructure and maintenance.

On-device option

Potential future option for reducing API dependency but not recommended as the initial cross-platform implementation.

13. Conclusion

Food Remedy can sustainably adopt AI without making AI a requirement for the product. The safest and most financially sustainable architecture is to keep all safety-critical and recommendation decisions deterministic and use AI only as an optional explanation and personalization layer.

For development and testing, Gemini's current Free Tier provides a practical way to experiment with the proposed functionality.

For a funded production implementation, Gemini 2.5 Flash-Lite is the recommended hosted service because its current token prices are substantially lower than the other compared providers and the expected workload is small.

Most importantly, if funding ends, Food Remedy should disable the AI layer rather than allowing the core recommendation system to fail. The deterministic engine and structured explanation templates should remain as the permanent baseline. This directly satisfies the sustainability requirement for RE016.

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